Gleb Tutubalin

Junior Software Engineer | Backend Developer | Developer Tools

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PROFILE

- Fourth-year BSc Software Development student at South East Technological University (SETU), Carlow, expected to graduate in 2027.
- Hands-on experience building software tools, database-backed applications, and web systems using C, Java, Python, JavaScript, and MySQL.
- Built C2obra, an AST-based branch coverage instrumentation tool for C (tree-sitter, custom C runtime library, Python automation, GitHub Actions CI) as a final-year project, benchmarked against TestCov and TestCoCa.
- Seeking a full-time junior or graduate role in software engineering, backend development, developer tooling, or test engineering.

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python, Go, JavaScript, PHP, Bash, x86_64 Assembly (NASM)
- **Databases:** MySQL, MariaDB, database schema design
- **Tools and Platforms:** Linux, Git, GitHub, GitHub Actions, GCC, Make, gcov/lcov, tree-sitter, StarUML
- **Web Technologies:** HTML, CSS, React, Angular, .NET, AJAX
- **Concepts:** Data Structures and Algorithms, OOP, MVC, DAO, service-layer architecture, AST parsing, branch coverage instrumentation, debugging, Agile/Scrum

EDUCATION

BSc in Software Development 2023 – 2027 (expected)

South East Technological University (SETU), Carlow, Ireland

- Year 1 GPA: 79.75% | Year 2 GPA: 77.58% | Year 3 GPA: 78.41%
- Relevant coursework: Software Engineering, Advanced Data Structures and Algorithms, Operating Systems, Cloud Development, Assembly and C, Systems Analysis and Testing, Object-Oriented Software Development, Web Programming and Databases, Computer Networks.

Higher Education Preparation Certificate (Professional English C1, NFQ Level 6) 2022 – 2023

South East Technological University (SETU), Carlow, Ireland | *GPA: 83.91%*

EXPERIENCE

IT Administrator June 2025 – August 2025

Megawatt LLC, Russia

- Rebuilt and optimised the company's internal network infrastructure from the ground up.
- Integrated cloud storage solutions and configured connections to external servers and network printers.
- Deployed and configured new computer systems, and configured and fixed employee account systems on office PCs.
- Supported network security and reliable connectivity across departments.
- Provided technical support and resolved hardware and software issues to maintain stable day-to-day operations.

Web Developer / IT Administrator June 2024 – August 2024

Lombard Companion LLC, Russia

- Developed and published the company website.
- Set up and maintained office IT infrastructure, including network connectivity, printers, internet access, and employee workstations.
- Supported the company's digital operations, assisted with basic online advertising setup, and resolved day-to-day technical issues.

PROJECTS

C2obra – Branch Coverage Instrumentation Tool for C 2026 – Present

Final-year project | *C, Python, Bash, Make, Linux, tree-sitter, gcov/lcov* | github.com/GlebTut/c2obra

- Developed a branch coverage instrumentation tool for C programs as a research-focused final-year project.
- Implemented AST-based parsing with tree-sitter to detect and instrument if, while, for, do-while, switch, and ternary branches.
- Built a Python pipeline that injects a custom C runtime library, records branch hit counters, runs Sikraken-generated test inputs in parallel, and produces HTML and CSV coverage reports.
- Set up a GitHub Actions CI/CD pipeline that runs the tool across benchmarks in parallel and publishes reports to GitHub Pages.
- Designed the system architecture and UML diagrams, and wrote supporting technical documentation.
- Tested on Test-Comp benchmark programs and evaluated against TestCoCa, TestCov, and Sikraken.

AutoSphere – Garage Management System 2024

PHP, JavaScript, CSS, MySQL, MVC, DAO | github.com/GlebTut/Garage_Management_System

- Worked in a team of five to build a garage management system with customer records, vehicle history, invoicing, and payment workflows.
- Implemented DAO and service-layer patterns and contributed to database schema design and optimisation.

Auction System – Desktop Application 2024

Java Swing, MySQL, MVC, DAO | github.com/GlebTut/Auction_System_Project

- Built a desktop auction application with user authentication, lot management, and bidding workflows.
- Designed a normalised relational database schema and implemented input validation and role-based access control.

x86_64 Assembly Projects 2025

NASM, Linux system calls, C | github.com/GlebTut

- Developed low-level x86_64 assembly programs using Linux system calls, covering process management and OS interaction.
- Applied knowledge of CPU architecture and memory management in low-level programming.

ADDITIONAL INFORMATION

- **Languages:** English (fluent), Russian (fluent)
- **Interests:** Judo (candidate for Master of Sport, with competition and coaching experience); currently training in MMA